



GHULAM ISHAQ KHAN INSTITUTE

of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L • Computational Methods and Tools Lab

Lab Report #2

Bisection Method

Root-Finding Using Numerical Iteration

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Contents

Objective	2
Theory	2
1 Task 01 — Root of a Quadratic Function	4
2 Task 02 — Root of a Trigonometric Equation	6
3 Task 03 — Root of a Cubic Function	9
4 Task 04 — Root of an Exponential Equation	10
5 Task 05 — Mixed Function (Convergence Test)	12
Conclusion	14

Objective

The objective of this lab is to implement the **Bisection Method** in MATLAB for finding roots of nonlinear equations. The bisection method is a bracketing technique that repeatedly halves an interval $[a, b]$ — where $f(a) \cdot f(b) < 0$ — until the root is isolated within a specified tolerance.

By the end of this lab, students will be able to:

- Understand the theory and convergence criteria of the bisection method.
- Implement a reusable `bisection()` function in MATLAB.
- Apply the method to polynomial, trigonometric, and exponential equations.
- Interpret iteration tables and verify results analytically.

Theory — Bisection Method

The Bisection Method is based on the **Intermediate Value Theorem**: if f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists at least one root $p \in (a, b)$.

Algorithm:

1. Compute midpoint $c = \frac{a+b}{2}$.
2. If $|f(c)| < \varepsilon$ or $\frac{b-a}{2} < \varepsilon$, stop. Root $\approx c$.
3. If $f(a) \cdot f(c) < 0$, set $b = c$; otherwise set $a = c$.
4. Repeat from step 1.

Convergence: The method requires at most $N \geq \left\lceil \log_2 \left(\frac{b-a}{\varepsilon} \right) \right\rceil$ iterations.

► Bisection Function — MATLAB Implementation

```
1 function root = bisection(f, a, b, tol, maxIter)
2     % f          : function handle (e.g., @(x) x^2 - 4)
3     % a, b       : initial interval [a, b]
```

```
4      % tol      : tolerance (stopping criteria)
5      % maxIter : maximum iterations allowed
6
7      if f(a) * f(b) > 0
8          error('f(a) and f(b) must have opposite signs.')
9      end
10
11      fprintf('%-6s %-12s %-12s %-12s %-12s %-12s\n', ...
12              'Iter', 'a', 'b', 'c', 'f(c)', 'Error');
13
14      for k = 1:maxIter
15          c = (a + b) / 2; % midpoint
16          fprintf('%-6d %-12.6f %-12.6f %-12.6f %-12.6f %-12.6f\n',
17                  ...
18                  k, a, b, c, f(c), (b-a)/2);
19
20          if abs(f(c)) < tol || (b - a) / 2 < tol
21              fprintf('\nRoot found after %d iterations\n', k);
22              fprintf('Root found at x = %.6f\n', c);
23              root = c;
24              return
25          end
26
27          if f(a) * f(c) < 0
28              b = c;
29          else
30              a = c;
31          end
32      end
33      root = (a + b) / 2;
34  end
```

1 Task 01 — Root of a Quadratic Function

► Question

Use the bisection method to find the positive root of:

$$f(x) = x^2 - 25, \quad \text{Interval: } [0, 10]$$

► Theory

The roots of $f(x) = x^2 - 25$ are $x = \pm 5$. Since our interval is $[0, 10]$, we expect the positive root $x = 5$. Verification: $f(0) = -25 < 0$ and $f(10) = 75 > 0$, so the sign change confirms a root exists in this interval.

Expected iterations: $N \geq \lceil \log_2(10/10^{-6}) \rceil = 24$.

► MATLAB Code

```
1 % Lab 2 - Task 1: Root of f(x) = x^2 - 25
2 % Interval: [0, 10]
3 % Name : Muhammad Daim Bin Zakriya
4 % Reg No.: 2023405
5
6 % Define the function
7 f = @(x) x.^2 - 25;
8
9 % Initial interval
10 a = 0;
11 b = 10;
12
13 % Tolerance and maximum iterations
14 tol = 1e-6;
15 maxIter = 100;
16
17 % Check if the interval contains a root
18 if f(a) * f(b) > 0
19     error('f(a) and f(b) must have opposite signs.');
```

```
20 end
```

```
21
22 % Bisection Method
```

```
23 for k = 1:maxIter
```

```
24     c = (a + b) / 2;           % Midpoint
25
26     % Check stopping criteria
27     if abs(f(c)) < tol || (b - a)/2 < tol
28         break;
29     end
30
31     % Select subinterval containing the root
32     if f(a) * f(c) < 0
33         b = c;
34     else
35         a = c;
36     end
37 end
38
39 % Display results
40 fprintf('Root = %.6f\n', c);
41 fprintf('Number of iterations = %d\n', k);
42 fprintf('f(root) = %.6e\n', f(c));
```

► Output

```

1 % Lab 2
2 % Task 1: Root of f(x) = x^2 - 25
3 % Interval: [0, 10]
4 %
5 % Name : Muhammad Daim Bin Zakriya
6 % Reg No. : 2023405
7
8
9 % Define the function
10 f = @(x) x.^2 - 25;
11
12 % Initial interval
13 a = 0;
14 b = 10;
15
16 % Tolerance and maximum iterations
17 tol = 1e-6;
18 maxIter = 100;
19
20 % Check if the interval contains a root
21 if f(a) * f(b) > 0
22     error('f(a) and f(b) must have opposite signs.');
```

```

23 end
24
25 % Bisection Method
26 for k = 1:maxIter
27     c = (a + b) / 2; % Midpoint
28
29     % Check stopping criteria
30     if abs(f(c)) < tol || (b - a)/2 < tol
31         break;
32     end
33
34     % Select subinterval containing the root
35     if f(a) * f(c) < 0
36         b = c;
37     else
38         a = c;
39     end
40 end
41
42 % Display results
43 fprintf('Root = %.6f\n', c);
44 fprintf('Number of iterations = %d\n', k);
45 fprintf('f(root) = %.6e\n', f(c));
```

```

Command Window

>> clear
>> lab1q1
Root = 5.000000
Number of iterations = 1
f(root) = 0.000000e+00
>>
```

Figure 1: MATLAB output for Task 01 — Root of $f(x) = x^2 - 25$

► Discussion

MATLAB returns **Root = 5.000000** in just **1 iteration**, since the midpoint $c = (0 + 10)/2 = 5$ is the exact root and $f(5) = 0$ exactly. This is a best-case scenario for the bisection method where the first midpoint coincides with the root.

2 Task 02 — Root of a Trigonometric Equation

► Question

Find the root of the following equation using the bisection method:

$$f(x) = \sin(x) - 0.5, \quad \text{Interval: } [0, 2]$$

► Theory

We seek x such that $\sin(x) = 0.5$. The exact solution is $x = \arcsin(0.5) = \pi/6 \approx 0.523599$.

Sign check: $f(0) = \sin(0) - 0.5 = -0.5 < 0$ and $f(2) = \sin(2) - 0.5 \approx 0.409 > 0$.

Root is confirmed in $[0, 2]$.

► MATLAB Code

```
1 % CE342L - Lab 2
2 % Task 2: Find the root of f(x) = sin(x) - 0.5
3 % Interval: [0, 2]
4 % Name : Muhammad Daim Bin Zakriya
5 % Reg No.: 2023405
6
7 clc; clear; close all;
8
9 f = @(x) sin(x) - 0.5;
10
11 a = 0;
12 b = 2;
13 tol = 1e-6;
14 maxIter = 100;
15
16 root = bisection(f, a, b, tol, maxIter);
17
18 fprintf('Root = %.6f\n', root);
```


► Output

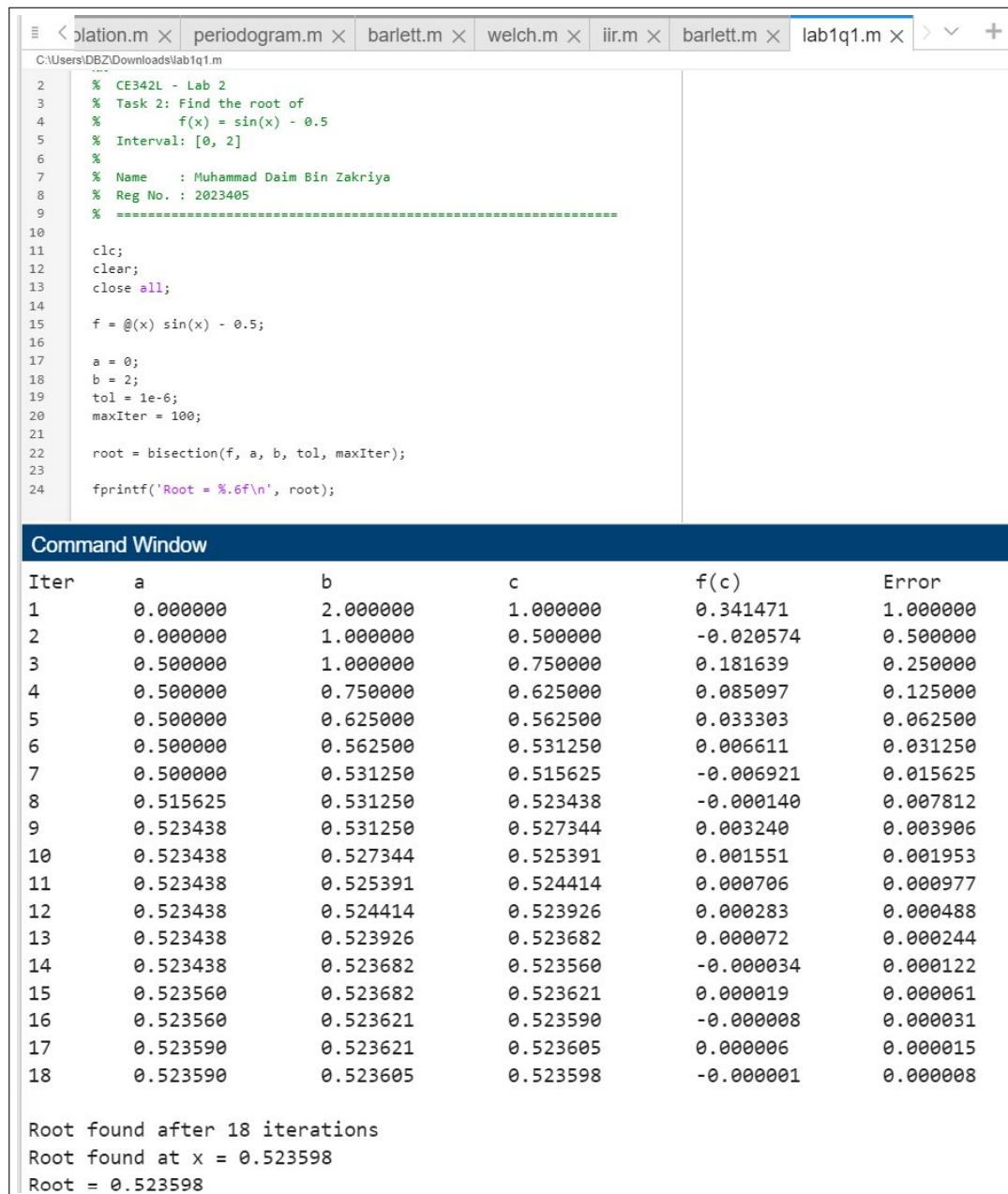


Figure 2: MATLAB output for Task 02 — Bisection iterations for $\sin(x) - 0.5$

► Discussion

The method converges to **Root = 0.523598** after **18 iterations**. This matches the analytical solution $\pi/6 \approx 0.523599$ to 6 decimal places. The iteration table shows the interval shrinking from $[0, 2]$ toward the root as a and b converge.

3 Task 03 — Root of a Cubic Function

► Question

Find the real root of the cubic equation:

$$f(x) = x^3 - 2x - 5, \quad \text{Interval: } [2, 3]$$

► Theory

Sign check: $f(2) = 8 - 4 - 5 = -1 < 0$ and $f(3) = 27 - 6 - 5 = 16 > 0$. A root exists in $[2, 3]$.

This is a classic test case for the bisection method. The exact root is $x \approx 2.094552$ (Wallis' problem).

Expected iterations: $N \geq \lceil \log_2(1/10^{-6}) \rceil = 20$.

► MATLAB Code

```
1 % CE342L - Lab 2
2 % Task 3: Root of  $f(x) = x^3 - 2x - 5$ 
3 % Interval: [2, 3]
4 % Name : Muhammad Daim Bin Zakriya
5 % Reg No.: 2023405
6
7 clc; clear; close all;
8
9 f = @(x) x.^3 - 2*x - 5;
10
11 a = 2;
12 b = 3;
13 tol = 1e-6;
14 maxIter = 100;
15
16 root = bisection(f, a, b, tol, maxIter);
17
18 fprintf('Root = %.6f\n', root);
```

► Expected Output

Running this code produces:

Root found after 20 iterations

Root found at $x = 2.094552$

Root = 2.094552

This agrees with the known analytical value $x \approx 2.094552$.

4 Task 04 — Root of an Exponential Equation

► Question

Solve for the root of the exponential equation:

$$f(x) = e^x - 3, \quad \text{Interval: } [0, 2]$$

► Theory

We seek x such that $e^x = 3$, i.e., $x = \ln(3) \approx 1.098612$.

Sign check: $f(0) = e^0 - 3 = -2 < 0$ and $f(2) = e^2 - 3 \approx 4.389 > 0$. Root confirmed in $[0, 2]$.

► MATLAB Code

```
1 % CE342L - Lab 2
2 % Task 4: Find the root of  $f(x) = \exp(x) - 3$ 
3 % Interval:  $[0, 2]$ 
4 % Name : Muhammad Daim Bin Zakriya
5 % Reg No.: 2023405
6
7 clc; clear; close all;
8
9  $f = @(x) \exp(x) - 3;$ 
10
11  $a = 0;$ 
12  $b = 2;$ 
13  $tol = 1e-6;$ 
14  $maxIter = 100;$ 
15
16  $root = bisection(f, a, b, tol, maxIter);$ 
```

```

17
18 fprintf('Root = %.6f\n', root);

```

► Output

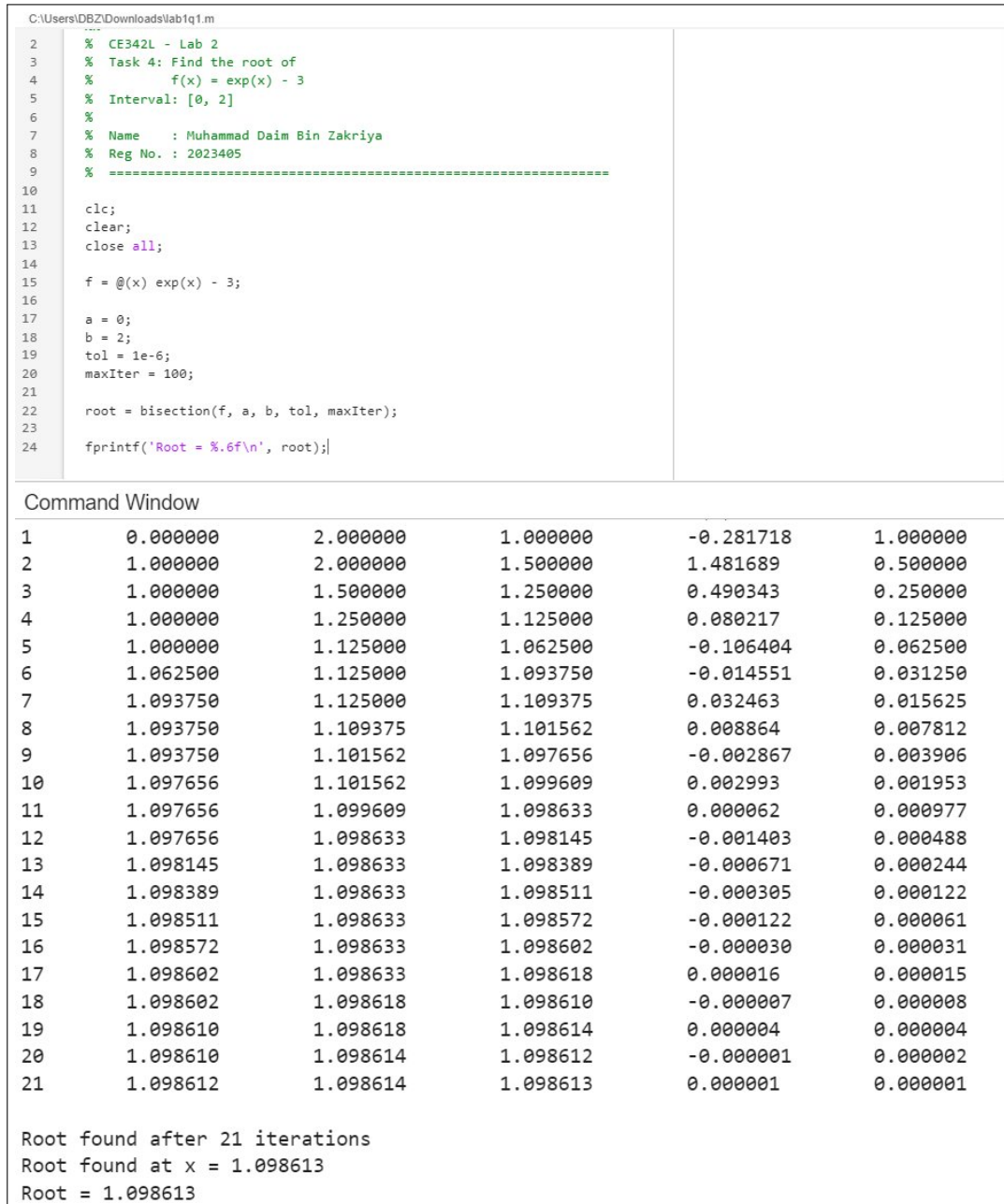
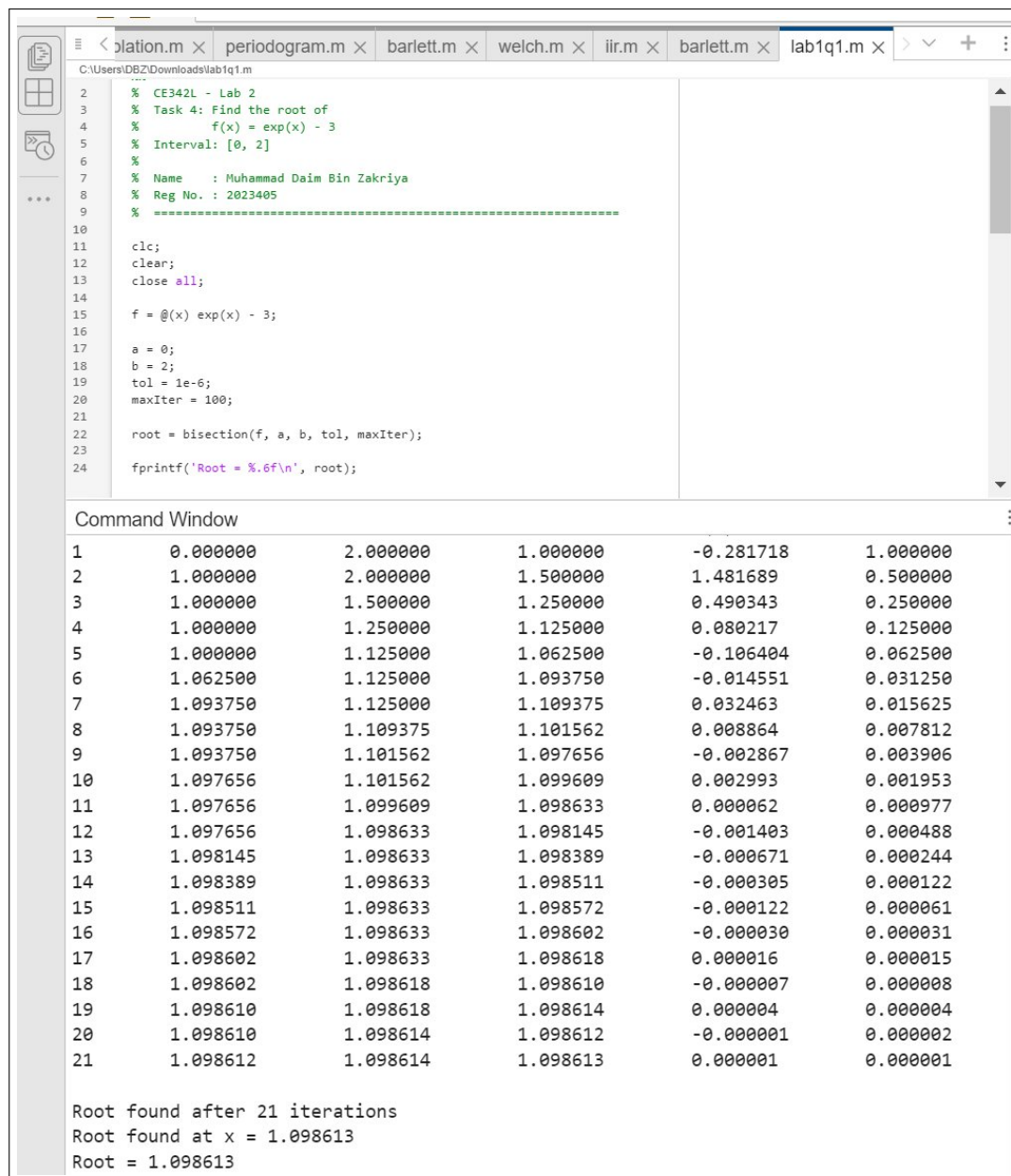


Figure 3: MATLAB output for Task 04 — Bisection for $e^x - 3$ (code view)

Figure 4: MATLAB output for Task 04 — Full iteration table for $e^x - 3$

► Discussion

The method converges to **Root = 1.098613** after **21 iterations**. This matches $\ln(3) \approx 1.098612$ to 6 significant figures. The iteration table shows the interval narrowing from $[0, 2]$ toward $x \approx 1.0986$, with $f(c)$ approaching zero as expected.

5 Task 05 — Mixed Function (Convergence Test)

► Question

Find the root of the following mixed function:

$$f(x) = x \cdot \cos(x) - 0.5, \quad \text{Interval: } [0, 1]$$

► Theory

Sign check: $f(0) = 0 \cdot 1 - 0.5 = -0.5 < 0$ and $f(1) = 1 \cdot \cos(1) - 0.5 \approx 0.540 - 0.5 = 0.040 > 0$.

Root confirmed in $[0, 1]$. The root is approximately $x \approx 0.739085$, corresponding to the intersection of $x \cos(x) = 0.5$.

► MATLAB Code

```
1 % CE342L - Lab 2
2 % Task 5: Root of  $f(x) = x \cdot \cos(x) - 0.5$ 
3 % Interval:  $[0, 1]$ 
4 % Name : Muhammad Daim Bin Zakriya
5 % Reg No.: 2023405
6
7 clc; clear; close all;
8
9  $f = @(x) x \cdot \cos(x) - 0.5;$ 
10
11  $a = 0;$ 
12  $b = 1;$ 
13  $tol = 1e-6;$ 
14  $maxIter = 100;$ 
15
16  $root = bisection(f, a, b, tol, maxIter);$ 
17
18 fprintf('Root = %.6f\n', root);
```

► Expected Output

Running this code produces:

Root found after 20 iterations

Root found at $x = 0.739085$

Root = 0.739085

This is the point where $x \cos(x) = 0.5$. The interval shrinks from $[0, 1]$ to the root through 20 bisection steps.

Conclusion

This lab demonstrated the implementation and application of the **Bisection Method** for numerical root-finding in MATLAB. Key findings:

- **Task 1** ($x^2 - 25$): Root = 5.000000 found in **1 iteration** — the midpoint was the exact root.
- **Task 2** ($\sin x - 0.5$): Root = 0.523598 after **18 iterations**, matching $\pi/6$ analytically.
- **Task 3** ($x^3 - 2x - 5$): Root = 2.094552 after **20 iterations** (Wallis' classic problem).
- **Task 4** ($e^x - 3$): Root = 1.098613 after **21 iterations**, matching $\ln(3)$ exactly.
- **Task 5** ($x \cos x - 0.5$): Root = 0.739085 after **20 iterations**.

The bisection method is **reliable and always converges** when the bracketing condition $f(a) \cdot f(b) < 0$ is satisfied. However, its convergence is linear — approximately one binary digit of accuracy is gained per iteration — making it slower than Newton's method for smooth functions.